

GridFlex AI

Complete Website Pages & Features Directory

Initiative: Yuva Yodha — Challenge 3
Target Domain: Renewable Integration & Local Grid Resilience

Target DISCOM: BESCO (Mahadevapura 33/11kV Substation)
Architecture: React 18 + Vite SPA & Python FastAPI

1. Overview & Evaluation Portals

Landing Page & National Energy Cockpit

Route: /

- Executive overview of software-defined grid resilience platform.
- Live national energy telemetry metrics (Solar MW, Wind MW, Grid Demand, Net Balance).
- Direct access persona quick-jump navigators (DISCOM Operator, System Evaluator, Microgrid Coordinator).
- Quick links to all 20 specialized operational modules and technical specifications.
- Bilingual interface toggle (English & Hindi) with instant instant reactivity.

Master About & 20-Page System Directory

Route: /about

- Comprehensive catalog of all 20+ specialized application modules.
- Data Provenance Guide: Verifiable classification between Real Data, ML Forecasts, and Simulation.
- End-to-End Technology Stack Matrix: Frontend (React 18 + Vite), Backend (FastAPI), Models (LightGBM, XGBoost, TreeSHAP).
- Role-based persona workflows for grid engineers, regulators, and innovation judges.
- Complete ISO 50001 & IEEE 1547 compliance mapping reference.

Judge Mode & Guided Storyboard

Route: /judge-mode

- 7-stage interactive evaluation tour demonstrating the complete Evening Deficit resolution.
- Guided walkthrough across NWP Weather -> ML Forecast -> Physics Twin -> MILP Solver -> P2P Clearing -> FLISR Healing.
- Automated step progression with timing controls and rubrics scorecard.
- Live verifiable telemetry audit snapshots and cost savings proof points (■15.25 Lakh saved).
- Built-in evaluator feedback and rubric criterion alignment.

Operator & Evaluator Authentication

Route: /login

- Instant 1-click login for 4 pre-configured demo roles (DISCOM Lead, Judge, Resilience Officer, Microgrid Lead).
- SQLite-backed authentication records audit ledger (/api/auth/records).
- Session token hydration with zero login friction for evaluators.
- Audit trail log showing timestamp, client IP, user ID, role, and authentication status.
- Public fallback access ensuring judges never get blocked during evaluation.

2. Real-Time Operations & Digital Twin

Live Autonomous Command & Dispatch Center

Route: [/command-center](#)

- Real-time SCADA telemetry display (64.8 MW Mahadevapura feeder load).
- Dynamic Energy Flow Sankey Diagram illustrating generation to load paths.
- Sub-cycle 120ms telemetry update tracking and non-disruptive gentle background polling.
- Manual 'Refresh Telemetry' control allowing on-demand updates without scroll reset.
- One-click 'Execute Dispatch Plan' button triggering automated setpoints to substation RTUs.
- Direct provenance badges marking real NWP weather and scaled NLDC load profiles.

Physics-Constrained Digital Twin Substation Simulator

Route: [/digital-twin](#)

- Interactive parameter sliders: Solar generation, wind speed, base demand, BESS readiness, flexible load enrollment.
- 4 Real-Day Analog Presets: Evening Ramp Deficit, Midday Solar Surge (+40%), Summer Heatwave, Monsoon Cloudburst.
- N-1 Contingency Drills: 1-click tripping of Substation BESS-01, North Solar Collector, or Feeder F-02.
- Real-time physics recomputation: Voltage profile (0.95-1.05 p.u.), transformer thermal loading (%), and net MW balance.
- Sub-cycle emergency mitigation response recommendations.

DISCOM Distribution Control Desk & Feeder Matrix

Route: [/discom](#)

- Substation-level monitoring for Mahadevapura 33/11kV primary distribution hub.
- Multi-feeder loading matrix (Feeder F-01 Tech Park, F-02 Residential, F-03 Industrial, F-04 Commercial).
- Feeder thermal stress alarms and phase unbalance monitoring.
- Automated tap-changer (OLTC) status and reactive power compensation tracking.
- Peak load curtailment and emergency reserve margin indicators.

Self-Healing Grid & Causal FLISR Engine

Route: [/self-healing](#)

- Fault Location, Isolation, and Service Restoration (FLISR) simulation.
- Pearl's Causal-Do Calculus engine evaluating counterfactual isolation actions.
- Interactive fault induction drill on distribution feeders.
- Automated sub-second tie-switch re-routing restoring power to healthy de-energized sections.
- Demand Response (DR) agent auction prioritizing critical hospital feeders.

3. Multi-Horizon Forecasting & Meteorology

Multi-Horizon Renewable Generation Forecast

Route: [/renewable-forecast](#)

- Multi-horizon day-ahead solar and wind generation forecast (24h/48h).
- P10, P50, and P90 confidence intervals generated with LightGBM quantile regression.
- Live meteorological telemetry: Global Horizontal Irradiance (GHI), DNI, ambient temperature, wind velocity.
- Curtailment risk probability assessment and weather alert banners.
- Exportable forecast dataset for dispatch scheduling.

Hourly Feeder Demand Forecast & Transformer Stress

Route: [/demand-forecast](#)

- XGBoost temporal regression model predicting 24h feeder demand curves.
- Evening duck-curve peak ramp detection (17:30 - 21:30 IST).
- Transformer thermal stress indicators and headroom utilization metrics.
- Historical vs forecast demand variance visualization.
- Predictive risk score for feeder overloading under extreme temperature.

SkyVision NWP Cloud Radar & Solar Irradiance Tracker

Route: [/sky-vision](#)

- High-resolution cloud vector radar tracking solar ramp events.
- Nowcasting solar drop prediction (0 - 120 minutes lead time).
- Real-time Global Horizontal Irradiance (GHI) heatmap over distribution feeders.
- Proactive BESS pre-charge triggers prior to cloud cover occlusion.
- Radar playback animation with speed and opacity controls.

4. Optimization, Flexibility & Micro-Markets

Flexibility Dispatch & Peak Shaving Solver

Route: [/flexibility](#)

- Mixed-Integer Linear Programming (MILP) algorithmic load shifting engine.
- Recharts Before vs After Demand Curve with glowing renewable supply area gradient.
- Coordinated Dispatch Schedule Chart: Hourly BESS charge/discharge & flexible load shift (MW).
- Scorecard: 9.8 MW peak reduction, 100% shortage elimination, ■1,42,500 DSM penalty avoidance.
- Automated operational directives dispatched to RTUs (commercial chiller pre-cooling, battery discharge).
- Interactive parameter sliders for BESS fleet readiness and flexible participation.

Peer-to-Peer (P2P) Micro-Market & Double Auction Ledger

Route: [/p2p](#)

- Continuous double-auction clearing engine matching local prosumer solar with consumer deficit.
- Recharts Market Clearing Dynamics Graph: Prosumer supply (kW), consumer bids (kW), cleared volume (kWh), and dynamic clearing rate (■/kWh).
- Standardized in Indian Rupees (■/kWh) anchored to Vidyut PRAVAH Day-Ahead Market pricing.
- CERC open access wheeling fee reconciliation (■0.85/kWh DISCOM fee).
- Cryptographic hash settlement ledger with audit timestamps and participant registry.
- Critical Load Protection Guarantee: 100% automated backup import for District General Hospital.

Virtual Community Storage (BESS) Fleet Manager

Route: [/storage](#)

- Multi-node 40 MWh distributed Battery Energy Storage System (BESS) fleet.
- Real-time State of Charge (SOC %), State of Health (SOH %), and temperature monitoring.
- State-space electrochemical degradation physics model (94% round-trip efficiency).
- Automated charge during solar peak (11:00-14:00) and discharge during evening peak (18:00-21:30).
- Individual node telemetry across Tech Park, Substation, Logistics, and Hospital battery units.

Dynamic Tariff & CERC DSM Arbitrage Engine

Route: [/tariff-engine](#)

- Time-of-Use (ToU) and Deviation Settlement Mechanism (DSM) price calculator.
- Zone-2 Western/Southern regional grid pricing blocks based on CERC DSM Regulations 2023.
- Automated battery arbitrage optimizer calculating profit margins (Buy low, Sell high).
- DSM over-drawal and under-injection penalty exposure model.
- Interactive arbitrage matrix ranking commercial profitability across time windows.

3D Topological Feeder Spatial Digital Twin

Route: [/spatial-twin](#)

- Geospatial visualization of 33kV and 11kV distribution network topology.
- Interactive transmission line congestion highlighting and power flow direction vectors.
- Substation node status indicators (Healthy, Warning, Critical) with voltage and loading metrics.
- Congestion, Voltage, and Flow view mode filters.
- Gentle background telemetry update (300s) preventing user scroll disruption.

5. Explainable AI, Copilot & Resilience Auditing

TreeSHAP Explainable AI Attribution Studio

Route: [/explainable-ai](#)

- Additive Shapley feature attribution waterfall charts verifying mathematical additivity.
- Quantitative decomposition of dispatch decisions into solar irradiance, load ramp, temperature, and SOC features.
- Natural-language operational rationale explaining exactly WHY specific setpoints were chosen over alternatives.
- Reflects ISO 50001 energy auditing standards and transparent non-black-box governance.
- Interactive feature toggle comparing unconstrained vs physics-constrained model outcomes.

Incident Co-Pilot & 15-Language RAG Voice Assistant

Route: [/incident-copilot](#)

- Domain-specific Retrieval-Augmented Generation (RAG) assistant for substation operators.
- Ingested knowledge base from CEA grid standards, IEEE 1547, and CERC operating codes.
- Multilingual voice speech-to-text input across 15 Indian & global languages.
- Emergency contingency guidance with zero external paid API key requirement.
- One-click voice recording with instant speech synthesis and transcript dispatch.

Autonomous AI Grid Copilot

Route: [/copilot](#)

- Full-page interactive conversational assistant for grid engineering and optimization.
- Pre-seeded prompt chips: 'Grid status summary', 'BESS dispatch now', 'Active fault report', 'P2P trade overview'.
- Grounded response generation with citations to indexed regulatory documents.
- Quick copy and export capabilities for operator shift logbooks.

Transparent 4-Pillar Resilience Index

Route: [/resilience](#)

- 100% transparent composite score (0 - 100) eliminating opaque black-box indices.
- Verified mathematical formula: $\text{Resilience} = 0.25 \times (\text{Renewable}) + 0.25 \times (\text{Headroom}) + 0.25 \times (\text{BESS SOC}) + 0.25 \times (\text{Flexibility})$.
- Unit-tested arithmetic with dynamic score contribution breakdown bar.
- Ranked prescriptive improvement directives with projected score uplift.
- Audit export functionality for regulatory compliance submissions.

System Architecture & Data Integrity Matrix

Route: [/architecture](#)

- End-to-end technical pipeline visualization from sensory inputs to RTU actuation.
- Full Data & Demo Integrity Disclosure Matrix classifying all telemetry sources.
- Documentation of zero-hardware software architecture for low-cost DISCOM deployment.
- Mathematical formulation references for MILP solver, TreeSHAP, and double auction.